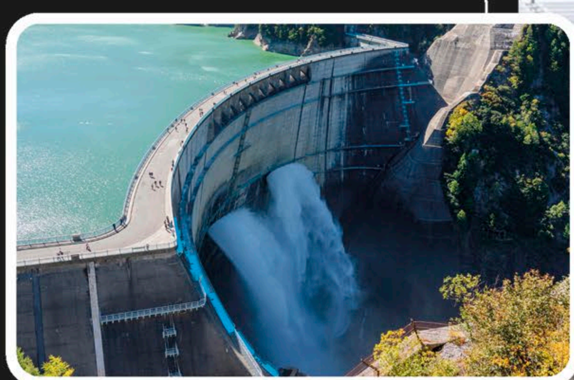
ENERGY STORAGE

BIG BATTERIES

One of the biggest challenges for the renewable energy industry is producing a constant supply when wind and sunshine cannot always be guaranteed. One solution is to store energy as it is generated for later use. Gigantic rechargeable batteries, similar to those used in household electronics and electric cars (see page 89), are one way of storing this energy to ensure a continuous power supply.

PUMP POWER

Hydroelectric power stations store water's potential energy. Water stored at the top of a slope powers turbines as it flows downhill. It can be pumped back uphill when energy demands are low and stored until needed to power the turbines again.



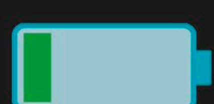
STORING ENERGY

Energy is the ability to make things happen. It comes in many different forms, including light and heat. Energy that is stored, ready to be used, is called potential energy. Energy can be stored in many ways—for example, by lifting heavy objects, by spinning objects, or as electric charge in a battery.

A machine called a flywheel stores kinetic (movement) energy.

A suspended weight stores gravitational potential energy.

A squashed spring stores elastic potential energy.



A rechargeable battery stores electrical energy.